

Namita Sardana

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Profile

I'm a data science graduate from RMIT, and what I enjoy most is using data to make sense of everyday things. I like the whole process of exploring a problem, digging into the details, and figuring out how to turn raw information into something useful; whether that's a simple visual, a recommendation, or just a clearer way to look at things. Over time, I've built experience across different areas, from developing machine learning models to working on more strategic programs in corporate settings. What I've realised is that I really enjoy the mix of logic and creativity, especially when it's applied to real-world problems that people actually care about. I learn best by doing, and I'm always looking for opportunities where I can grow, contribute, and work with people who care about making a difference through better tools, better insights, or even just better questions.

Experience

Curious Cat (Melbourne, Australia) | Data Science Intern

July 2025 – November 2025

As part of my final-semester industry internship, worked with Curious Cat, an analytics division of Arrotex Pharmaceuticals, on a project called VetIQ, designed to help veterinary clinics better understand and care for their clients and patients. The goal was to improve things like client loyalty, day-to-day efficiency, and personalised care through tools that highlight risks, spot opportunities, predict future trends, and track patient wellness; all presented in a unified dashboard. Within the clinic-focused team, ideated and led the development of the "Opportunity Radar," which analysed patterns in the services clients use. This helped uncover which services are often linked together and where clinics might be missing opportunities to offer additional care. For instance, it revealed common patterns like clients who come in for consultations often needing medication, or preventive care leading to surgical needs. These insights were turned into simple, practical recommendations for clinics. Also contributed to the Client Retention Tracker, improving how the system identifies clients who may stop returning. This involved strengthening key signals like how recently and how often clients visit, and how much they spend. The work helped clinics better spot at-risk clients and take action early. Beyond these features, supported the overall process of preparing data, building and updating the dashboard, and setting up automated workflows so everything runs smoothly.

Focus Bear (Melbourne, Australia) | AI Engineer Intern

November 2024 – March 2025

Interned at Focus Bear, an AI-powered focus enhancement app designed to support adults with ADHD and autism. Worked on building a coarse filter feature for the Focus Bear app, designed to help block distracting websites by quickly deciding whether a site is relevant to what a user is trying to focus on. A key goal was to reduce reliance on GPT-4o, which helped cut API costs and improve response times. Started with simple approaches like using regex (pattern matching) to pick up keywords, then gradually improved the feature to better understand the meaning behind a user's goal (like studying or working) and the content of a website. This later evolved to include BERT, which allowed the feature to understand context and meaning more effectively, rather than just matching words. The final version compared a user's intention with website information and made a clear decision using a combination of methods. This led to a noticeable improvement in accuracy (about 5.5%) and made the feature more reliable, even when dealing with new or unfamiliar websites. Along the way, worked through challenges like deciding how strict the filter should be, handling edge cases, and balancing accuracy with speed and cost. Overall, this experience built a strong understanding of how to create practical, real-world AI features, especially in a sensitive space like helping people stay focused and reduce distractions.

McDonald's (Melbourne, Australia) | Crew

September 2024 – Present

Working at McDonald's has helped me build strong communication, teamwork, leadership, and problem-solving skills in a fast-paced customer service environment. I interact with customers daily, handle orders, and resolve issues efficiently while maintaining a positive attitude under pressure. The role has strengthened my ability to think on my feet, multitask effectively, and adapt quickly during busy periods. I work closely with a diverse team to ensure

smooth day-to-day operations and a positive customer experience, while also supporting team coordination and assisting newer staff members when needed.

IIT Kharagpur – Entrepreneurship Cell (Remote) | Machine Learning Intern

May 2020 – July 2020

Focused on building and improving AI tools for real-world use, particularly in healthcare and image-based applications. A key project involved developing a model that could analyse surveillance camera footage to detect whether people were wearing face masks, an important need during the COVID-19 pandemic. Worked on taking an initial version of the model and making it more accurate and reliable. This involved improving how it learned from images using convolutional neural networks (a type of neural network designed to understand visual patterns), helping it better recognise faces and correctly identify whether a mask was being worn, even in challenging conditions like different lighting, angles, or partial obstructions. Through a series of refinements and testing, improved the model's overall accuracy by more than 9% compared to earlier versions. This made it more dependable for real-world use, where consistency and precision are critical.

Education

RMIT University, Master of Data Science
Melbourne, Australia

February 2024 – December 2025

SRM University, Bachelor of Computer Science & Engineering
Delhi-NCR, India

July 2019 – May 2023

Projects

T20 Match Engine (Sports Analytics Framework)

Developed a unified, 360-degree analytics engine designed to evaluate T20 cricket matches ball-by-ball across both innings from both batting and fielding perspectives. Recognising that the two halves of a match serve distinct strategic purposes, built a dual-model architecture: an XGBoost regressor for the first innings to predict Expected Final Score (eScore) based on match state, venue conditions, and scoring tempo, paired with an XGBoost classifier for the second innings to continuously estimate live Win Probability (WP) during a chase. From these core models, derived advanced situational metrics including Runs Above Expectation (RAE) for Innings 1 batters, Expected Runs Saved (ERS) for fielders, and Win Probability Added (WPA) for Innings 2 players, alongside a dynamic Leverage Index (LI) that quantifies game pressure before every delivery. Built additional context-aware features like a Clutch Score to weight high-pressure contributions and a Pressure Index to measure the compounding impact of dot balls. This framework eliminates reliance on raw, context-blind statistics by supporting context-adjusted player rankings, identifying game-defining deliveries, measuring phase-by-phase team efficiency across powerplay, middle, and death overs, and rendering real-time dashboard visualisations that transition smoothly from target projection to live win probability. Beyond cricket, the underlying architecture is directly reproducible for any dual-phase or possession-based sport; for instance, in basketball (NBL/NBA), the same dual-engine logic translates seamlessly by swapping cricket's target-setting and run-chase phases for time-decaying expected possession value (ePoints) and real-time win probability, mapping Leverage Index directly to high-stakes possession management, late-game timeout decisions, and clutch player evaluation.

Eunoia: RAG Chatbot for Mental Health Support

As a part of my uni group project, Eunoia started as a way to explore how conversational AI could support mental wellbeing without crossing into unsafe or clinical territory. We wanted to build something that doesn't just talk about mental health, but listens responsibly. The system uses a Retrieval Augmented Generation (RAG) pipeline powered entirely by Ollama, employing "nomic-embed-text" model for semantic embeddings and "mistral-latest" model for retrieval-based generation. These embeddings are indexed using FAISS, enabling Eunoia to interpret nuanced, context-rich queries like "I feel burnt out but not sad" and return semantically aligned, evidence-based responses from verified mental health resources. The knowledge base combines a curated collection of psychological guides and coping resources with a questions.csv file containing conversational examples and human-aligned responses. This dual structure improved contextual precision by over 5% compared to baseline retrieval systems. A feedback-driven refinement loop continuously evaluates responses based on user feedback,

enhancing retrieval accuracy by about 8% and helpfulness by 2% over similar open-source mental health chatbots. Testers described Eunoia's tone as "reassuring yet factual", an intentional balance between empathy and accuracy. Ethical and emotional safety were central to design; if the model's confidence drops below a threshold, it activates a safety fallback that redirects users to trusted helplines like Lifeline or Headspace instead of hallucinating uncertain advice. Eunoia ultimately blends technology and empathy, a conversational companion that supports, informs, and respects the user's wellbeing boundaries while maintaining transparency and care.

CarbonWise: Personal Environmental Impact Tracker

This project started as a way to understand how much carbon we emit just by getting around every day. I originally planned to use my Google Maps Takeout data to track my own travel history, but since my location tracking was off, I ended up simulating 23 years of travel instead. From childhood outings and school bus rides to uni commutes and part-time job shifts, I mapped it all out. The dashboard shows total CO₂ emissions, distance travelled, breaks it down by transport mode (with flights separated), and even compares monthly emissions against a personal goal. It also suggests greener travel options and shows how much carbon you could save by switching, thanks to a decision tree model that figures out when walking, cycling, or public transport might actually make sense based on the trip type, distance, and potential impact. There's also a section where anyone can add their own trips and get similar insights instantly. Overall, it's a tool for reflecting on how we move through the world, and how we might do it a bit more sustainably.

Context-Aware Movie Recommendation Engine

Developed a content-based movie recommendation system that suggests films by analysing attributes such as genre, cast, and descriptions. The system incorporates additional contextual factors to enhance recommendations, including seasonal trends (e.g., horror movies in October, rom-coms near Valentine's Day), viewing preferences (watching alone, with family, friends, or a partner), and a "Weekend Mode" for binge-worthy content on weekends and lighter recommendations for weekdays. It also includes time-based filters like "Short Watch" mode for movies under 90 minutes and "Long Movie Night" mode for epic-length films. By leveraging these contextual elements, the system provides more relevant and personalised movie suggestions tailored to user needs.

Skills & abilities

○ Technical Skills:

- **Data Science & Machine Learning:** Experienced in end-to-end ML workflows including data preprocessing, feature engineering, model training, optimisation, evaluation, and experiment tracking. Proficient in Python, R, and libraries such as Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, XGBoost, and ML experimentation tools (MLflow / W&B).
- **Big Data & Distributed Computing:** Hands-on experience processing large datasets using Apache Spark, PySpark, and Hadoop-based systems. Skilled in building scalable analytics solutions on AWS EMR, and leveraging Google Cloud Platform for ML workloads.
- **Cloud, MLOps & Deployment:** Familiar with designing cloud-based ML workflows, integrating feature stores, managing model versioning, and deployment fundamentals. Experience working with AWS and GCP environments, Microsoft Azure fundamentals (Azure Storage, Azure AI services, and deployment basics), containerisation basics, and maintaining reproducible end-to-end ML pipelines.
- **AI, NLP & RAG Systems:** Experienced in developing AI applications across NLP, embeddings, transformers, and Retrieval-Augmented Generation (RAG) systems. Hands-on experience with FAISS vector indexing, semantic search, embedding pipelines, Ollama-based local LLM workflows, prompt engineering, retrieval evaluation, and knowledge-grounded response generation.
- **Data Visualisation:** Skilled in creating dashboards and visual analytics using Tableau, Power BI, Matplotlib, Seaborn, and R visualisation tools.
- **Statistical Analysis:** Strong foundation in statistical modelling, A/B testing, hypothesis testing, and exploratory data analysis (EDA) using Python, R, and SAS.
- **Databases & APIs:** Proficient in working with SQL databases and NoSQL (MongoDB). Experienced in developing and integrating REST APIs for ML applications using Flask and FastAPI.
- **Automation & Workflow Orchestration:** Experienced with building automated data and model workflows using n8n, Git/GitHub workflows, and custom integrations to streamline analytics processes.
- **Web Application Development (Basic):** Experience creating interactive data products and ML dashboards using Flask and Streamlit.

- **Project & Collaboration Tools:** Proficient with Trello, Git/GitHub, and Overleaf for project management, documentation, version control, and collaborative work.
- **Soft Skills:**
 - **Adept Communication:** Communicate complex technical insights clearly and effectively to technical and non-technical stakeholders.
 - **Problem Solving:** Analytical and systematic approach to identifying root causes and designing data-driven solutions.
 - **Time Management:** Skilled in managing multiple ongoing tasks while meeting deadlines efficiently.
 - **Relationship Building:** Strong interpersonal skills, fostering collaboration with teammates, clients, and cross-functional partners.
 - **Adaptability:** Quick to learn and adapt to new tools, methodologies, and project requirements.
 - **Decision Making:** Make thoughtful, data-informed decisions considering both short-term impact and long-term goals.

Certifications

- **Google Data Analytics Specialisation:** Completed the Google Data Analytics Specialisation, gaining foundational knowledge and skills in data analysis. Covered essential concepts such as data cleaning, data visualisation, statistical analysis, and the use of tools like spreadsheets, SQL, and Tableau. Strengthened the ability to analyse datasets, derive meaningful insights, and communicate findings effectively.
- **30 Days of Google Cloud:** Successfully completed the "30 Days of Google Cloud" certification, acquiring hands-on experience with various Google Cloud services, including Google Cloud Storage, BigQuery, and Google Compute Engine. Enhanced understanding of cloud technologies and their applications in data science, enabling efficient handling of large-scale data processing tasks and cloud-based resource management.

Extracurriculars

- **Willis Towers Watson (Melbourne, Australia), Micro-internship:** Participated in a micro-internship at Willis Towers Watson, focusing on analysing and improving the company's Employee Assistance Program (EAP) by introducing data-driven insights. Worked as part of a team to evaluate the existing scenario of employee well-being and proposed optimisations using data analysis. Suggested enhancements to support mental health initiatives and improve the overall workplace environment. This experience developed strong data analysis and problem-solving skills in a business context emphasising workplace well-being.
- **AIA (Ho Chi Minh City, Vietnam), Micro-internship:** Selected as one of 20 students from RMIT to participate in a major industry challenge hosted by AIA Insurance in Ho Chi Minh City, Vietnam. The challenge focused on developing innovative solutions related to health and well-being services. Worked in a team of diverse individuals to identify ways to improve customer engagement and create better strategies for addressing health concerns. Through brainstorming, data analysis, and solution development, successfully proposed actionable recommendations presented to senior AIA staff. This experience refined teamwork, communication, and problem-solving skills in an international and industry-specific setting.
- **Vice President, Computer Society of India (CSI):** Led initiatives to promote computer science education and awareness within the community. Organised events, workshops, and technical sessions for students.
- **Hackathons & Tech Events:** Actively participated in and contributed to organising hackathons, technical talks, and collaborative coding challenges at university and community levels.

Working Rights

Temporary Graduate Visa (Subclass 485) with full, unrestricted working rights.

References

- **Mike Da Gama, VetIQ**
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- **Jeremy Nagel, Focus Bear**
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